AWS Competency Program

Demystifying APN Service Delivery Program and Reviews for prospective AWS DevOps Competency partners

Amazon Web Services

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DevOps Consulting Competency Technical Controls Calibration Guide

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Tech Controls Calibration Guide

- Case study example qualification
- Case study examples
- Deep-dive for every technical control:
  - Why is the control important?
  - How can the control be implemented?
  - What a good example looks like?

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DevOps Customer Example Qualification

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Case Study Qualification

- Transforms customer's DevOps practice/approach to AWS Cloud-based infrastructure as code.
- Leverages AWS compute, storage, or network services and essential cloud management toolsets

AWS CodeStar
AWS CodePipeline
AWS CodeBuild
AWS CodeCommit
AWS CodeDeploy
AWS OpsWorks
AWS CloudFormation
AWS Config
AWS Systems Manager
AWS Elastic Beanstalk
AWS Lambda
Amazon Elastic Container Service (Amazon ECS)
AWS X-Ray

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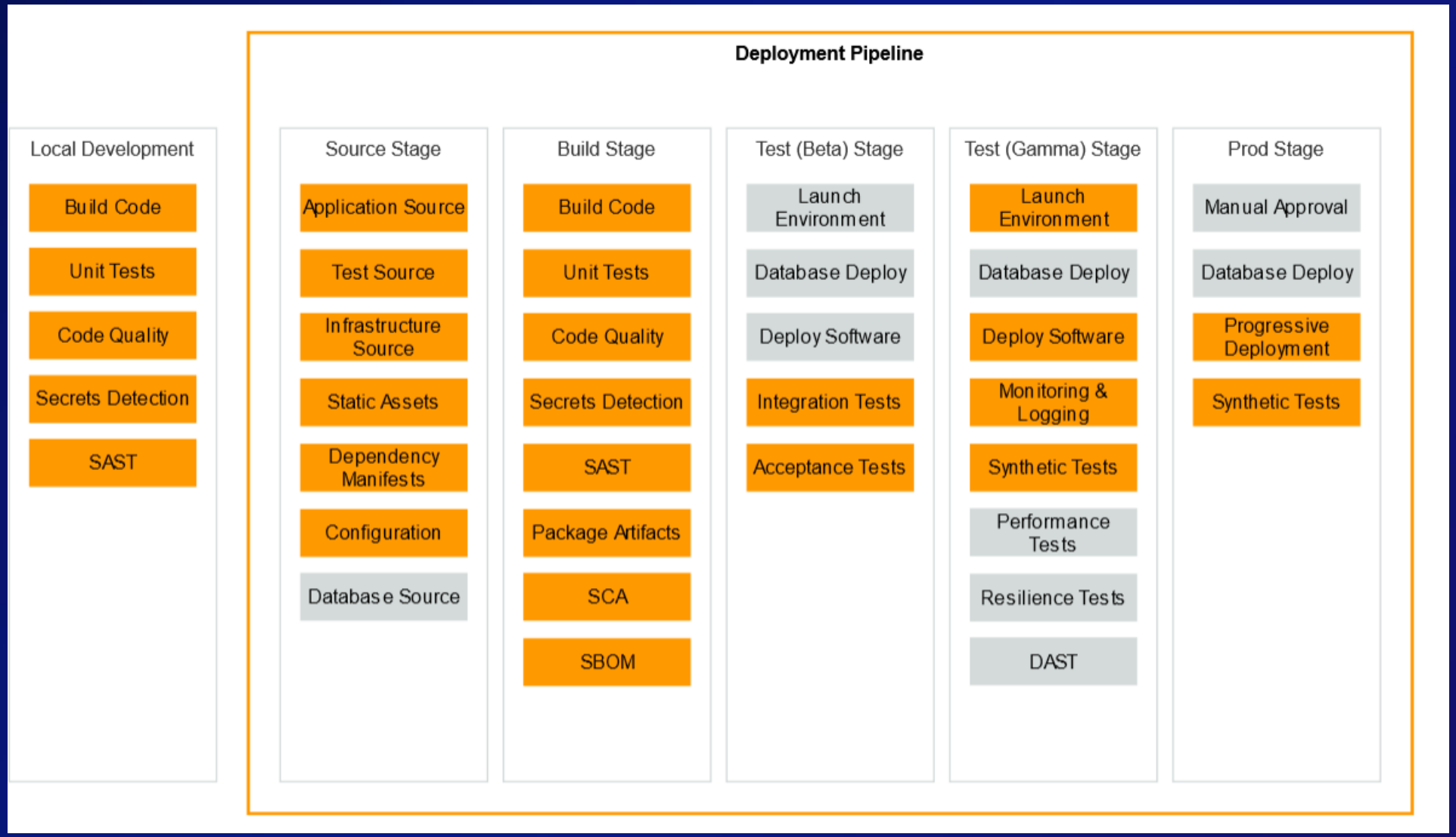
Deployment Pipeline Reference Architecture



https://pipelines.devops.aws.dev/application-pipeline/

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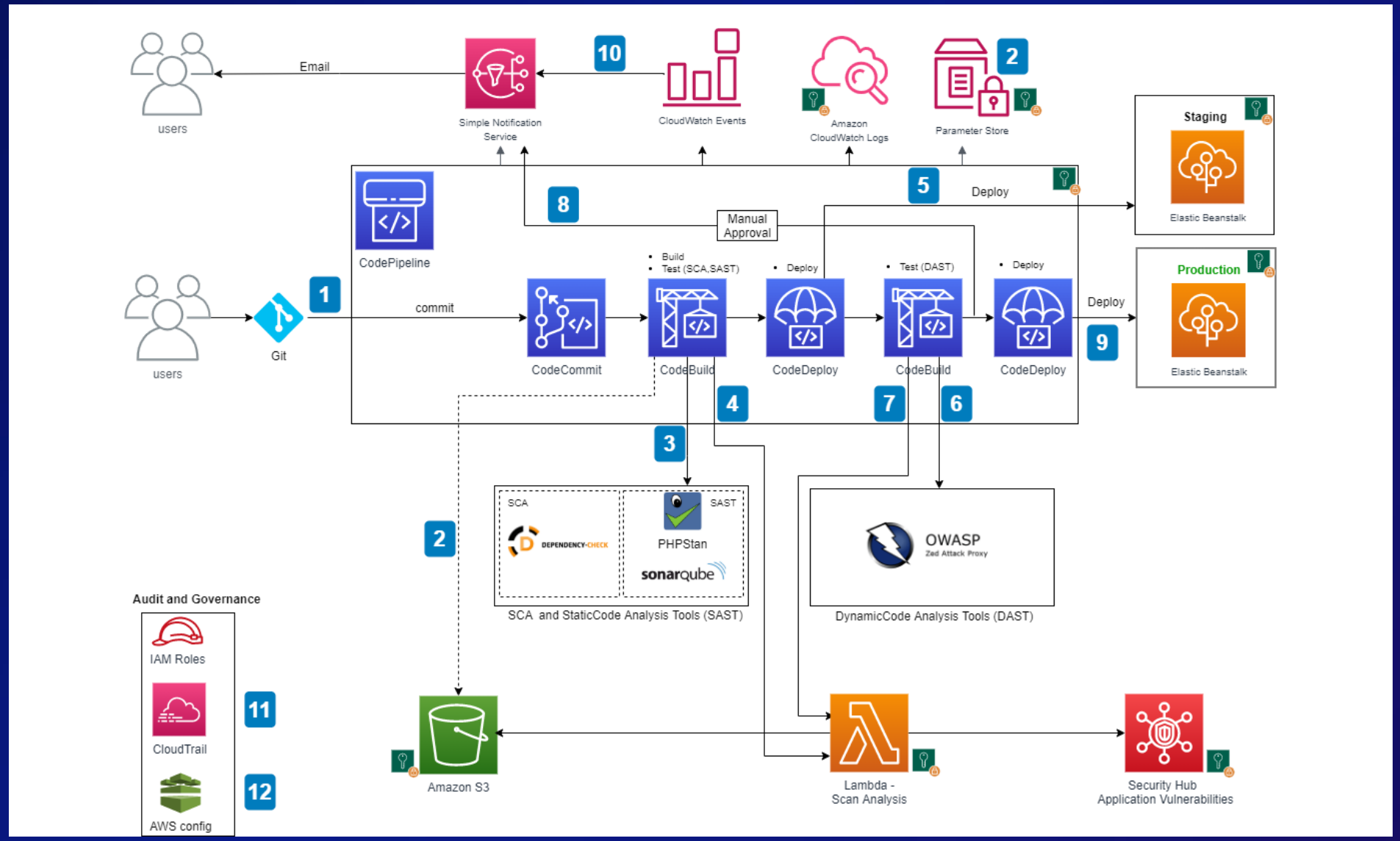
DevSecOps Example Architecture



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Common Issues With Applications

- Case studies
  - Are not DevOps *business* problems
  - Solution architecture does not describe the pipeline
  - Metrics do not tie back to the problem
- Architecture diagrams

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Case study: problem statement

Case Study	What does good look like	Comments
Problem Statement	The application does not handle spikes in user traffic and leads to 3-4 outages every month. This resulted in poor customer experience and loss in revenue.	• Is a problem statement • Focuses one issue - Scalability • Provides baseline data points • Talks about business impact

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Case study: proposed solution

Case Study	What does good look like	Comments
Proposed Solution & Architecture:	Standardize and automate infrastructure creation with infrastructure as code using Terraform. All infrastructure changes go through a CI/CD pipeline that applies quality, security and policy checks on any infrastructure changes. Automated the AMI creation for all the 30 applications that make up the website. We were then able to implement auto scaling using the "Golden AMIs". Auto-scaling was configured with a dynamic scaling policy that uses the Amazon CloudWatch CPUUtilization metric to add capacity when the instance exceeds 90 percent utilization for a specified length of time.	• Highlights devOps capabilities of IaC and CI/CD • Proposed solution addresses the problem statement • Quantifies using data points • Highlights your deep understanding of AWS services like auto scaling groups, monitoring with Amazon Cloudwatch, etc… • Mentions DevOps tools used (Note: the tools can be non-AWS services)

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Metrics for success

Case Study	What does good look like	Comments
Metrics for Success:	On an average there were 2-3 issues with every release related to configuration differences between prod and non-prod. All the non-prod and prod environments were standardized by IaC. This led to zero production issues related to misconfigurations. Implemented auto-scaling with a dynamic scaling policy that uses the Amazon CloudWatch CPUUtilization metric to add capacity when the instance exceeds 90 percent utilization for a specified length of time. The auto-scaling capabilities of the infrastructure led to a reduction of cloud spend by 200K per month as compute is dynamically allocated based on customer demand.	• Focuses on metrics and offers data points highlighting previous state and current state. • Addresses the problem statement • Highlights additional benefits from the implementation (all infrastructure is now standardized and automated)

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DevOps Practice Requirements

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Overview of DevOps Practice Requirements

Organization Change

Infrastructure as code

Continuous Integration/Continuous Delivery CI/CD

Monitoring & logging, version controls, security, etc.

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Organizational Change

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PROC-001

Customer Assessment of Internal Organization

- Align assessment approach with DevOps principles
- Demonstrate a well established criteria for assessment
- Provide organizational recommendations for DevOps approaches

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PROC-001

How can you implement this?

Structure the evidence as you see here!

- Customer Assessment Area and criteria
  - Structure, process and culture
- Customer DevOps practice gaps
  - Agility, empowerment
  - Automating and tooling
  - Cross-functional integration
- Actionable recommendations deveopment

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PROC-001

Good Example vs Unacceptable/Insufficient Information example

CLICK HERE FOR THE FULL EXAMPLE!

"AnyCompany has an internal methodology to help transforming organizations to DevOps (partner attached documents/links). we have defined a process to assess organizations and understand the actual state (as-is), diagnose their process, tools and teams to present future state (to-be) with the principal objective of change the way that the company develop, test, protect and deploy its software. This assessment classified organizations in maturity categories (reactive-proactive-predictive) and levels (baseline - repeatable - consistent - managed - optimized) based on 3 aspects: Culture, process and tech. This way can define and action plan with specific roadmap for working hand by hand with them..."

"Our DevOps projects are defined based on the specific needs of the client. We provide bespoke solutions and our project risk assessment based on customers' organizational strength and pain points."

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PROC-002

Methodology for Organizational Change Recommendations

- Establish a process to recommend changes for various departments
- Describe desired outcome of the recommended changes
- Develop initiatives/trainings required for DevOps transformation

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PROC-002

Good Example vs Unacceptable/Insufficient Information example

*"AnyCompany has a Change Management practices as a framework on the delivery of Dev Ops projects (**partner attached documents**). This framework addresses changes at organizational, program and individual level and ensures various stakeholder teams are aligned, trained/empowered thereby accelerating business adoption.*

*Additionally, we work with customer to **build with the client a Change management process** during the development life cycle based on assessment of customer's internal organization. This helps to identify the **scope** of a change initiative, and **a single threaded owner** to align cross-functional **resources and champions** of the change objective. Ultimately, it led to organization efficiency with DevOps projects."*

"We have a change management best practice for all customers and developed a communication plan for change management (generic, not specific to DevOps)"

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Infrastructure as Code

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PRIAC-001

Templated Infrastructure Provisioning

- Create and manage AWS resources via repeatable deployments
- Use templated infrastructure provisioning (AWS CloudFormation Terraform, etc.)
- Leverage **AWS CloudFormation Templates**

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PRIAC-001

Good Example vs Unacceptable/Insufficient Information example

"AnyCompany has a repository of CloudFormation templates for our AWS Dev&Ops customer engagement (attached examples). As a DevOps best practice, every cloud project has to provision IT resources as code from a centralized repository (Infrastructure as Code Reuse Library). This has repeatable and reusable solutions from developer community. We have a variety of resources for systems administrators and developers to deploy to AWS, including a knowledge center, Online Infrastructure as code Community, user forum and wiki. For every business case of our customers, we have separated Dev, Test and Production environments. Please see examples of CloudFormation templates for the four submitted customer case studies. "

"We use templated infrastructure provisioning and customize if needed for the specific project" (no example templates)

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PRIAC-002

Configuration Management

- Establish a process for automatic configuration management
- Identify configuration items and desired state, review & monitor
- Leverage AWS configuration management tools such as AWS Systems Manager, AWS Config or third-party tools

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PRIAC-002

Good Example vs Unacceptable/Insufficient Information example

*"AnyCompany has developed an automated process for configuration management. Automated configuration management practices have been deployed in the customer examples X, Y. We use **AWS Systems Manager** to collect software inventory data, apply OS Patches and run maintenance tasks using SSM Automation Documents. The basic configuration of the Systems Manager is **deployed via Landing Zone / Control Tower**. It deploys CloudFormation Templates which sets up the resource data sync to a central account / bucket, **maintenance windows for automated Instance patching via Instance Tags.** The **attached document** is what we provide to the client about best practices and AWS tooling for guiding how to manage configurations and infrastructure updates in our Configuration Management Practices. "*

"We use configuration management such as applying OS patches for all DevOps project" (no documented process or technical demonstration)

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PRIAC-003

Policy as Code

- Establish policy as code process to enforce compliance and remediation
- Continuously monitor throughout software delivery lifecycles
- Leverage Policy as code tools such as AWS Config Rules, CloudFormation Guard, and others

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PRIAC-003

Good Example vs Unacceptable/Insufficient Information example

*"AnyCompany has mature process for policy as a code where we use **AWS Config** to set up and **monitor the compliance status** of AWS resources. Specifically, we set up AWS Config rules for **automatic remediations** for violations like unencrypted S3 Buckets or unrestricted Remote Access in place. We send the **configuration snapshots to a central log** archive account which it can be further analyzed. The account owner takes further steps to review **and fix the found config violations** if needed. Additionally, we use Landing Zone / Control Tower to rollout things like Config and CloudTrail to all account and use SCPs to ensure that those settings can't be altered or disabled.*

*We share the best practices on how to set up and use AWS Config with our clients. See attached document and screenshots on how AWS Config is implemented. (**Partner attached documents**)."*

"We use AWS config to manage policies, we set up automated policy evaluation and remediate actions for DevOps project" (no example or templates)

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Continuous Integration/Continuous Delivery CI/CD

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PRCICD-001

Software Release Workflows

- Establish a release process model to build, test, and deploy code
- Identify deployment (blue/green, canary, or hybrid) and rollback strategy
- Leverage tools such as AWS CodePipeline & resources on DevOps and AWS

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PRCICD-001

Good Example vs Unacceptable/Insufficient Information example

*"AnyCompany has a defined release management process (**see attached documents**) to manage the software lifecycle transition from development to launching new release with speed and reliability. We developed an internal pipeline workflow management capability using **AWS CodePipeline and CodeCommit** to facilitate Continuous Integration and Continuous Delivery (CI/CD). See **attached release workflow example** and backlog release management metrics provided to customer. "*

We used a blue-green deployment across all our projects and selected the right release workflow based on customer requirements. (no examples)

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PRCICD-002

Build and Test Code

- Establish a process to compile code, run tests and produce software packages ready to deploy

- Enable early bug detection and ensure code quality

- Leverage tools such as AWS CodeBuild with CodePipeline, Jenkins or CodeCov

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PRCICD-002

Good Example vs Unacceptable/Insufficient Information example

“AnyCompany’s DevOps best practice is based on "Infrastructure As Code" (Ansible and CDK) to compile, test and deploy software components automatically. Our core Continuous Integration and Continuous Delivery (CI/CD) processes have the automation to build new development when merging code in the source control repository. The ***source code for automation (Jenkins pipelines, Ansible, CDK)*** *is stored in "git" versioned. Also, this process triggers* ***quality assurance with automated test*** *like* ***static code review, security code review, unit test****, among other testing practices. These ensures we are delivering the best possible customer experience for their DevOps workloads. Attached is a document of how the build & test automation framework is presented and applied to customer example. “*

We used a AWS managed services to build and test code and ensure code quality meets customer requirements.

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Version Control

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PRVCL-001

Source Control

- Set up source control systems for customers transforming to DevOps approaches

- Demonstrate expertise and enable customers in AWS CodeCommit, GitHub, Bitbucket, or others

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PRVCL-001

Good Example vs Unacceptable/Insufficient Information example

"The AnyCompany has source control management as a practice for all IT implementations. We use a variety of source control systems including Bitbicket, GitHub as wells as CodeCommit. Project teams maintains these resources in the version control model and educate customers with each asset update. This allows customers to track changes over time, reducing risk while providing continuity and flexibility. We provided a document on how to set up source control systems including AWS CodeCommit repository according to customers' preferences in terms of technologies. We recommend best practice, branching strategy and tooling based on customer development methodology and product strategy. (partner attached document link)"

We implement version control for all DevOps projects when working with customers.

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Microservices

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PRMSC-001 – Container Services/Serverless Computing

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PRMSC-001

Container Services/Serverless Computing

- Define strategy for containerization or serverless computing architectures
- Establish methodologies to leverage Amazon ECS, EKS or Lambda for applicable use cases
- Communicate benefit of microservices to customers' DevOps transformation

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PRMSC-001

Good Example vs Unacceptable/Insufficient Information example

"…At AnyCompany we have defined different architecture patterns, including event-driven ***and microservices strategies to consume serverless components*** *like API Gateway, Lambda, Event Bridge, DynamoDB, SQS, AWS Fargate and others. For containerized applications, we have a defined architecture pattern using AWS EKS and AWS ECS container orchestration services. We* ***recommend containerized application when portability and full control of the environment is a high priority****, especially for customers with migrating legacy applications to the cloud. Both approaches are not mutually exclusive, they can complement each other. For instance, …*

We leverage microservices whenever possible for customer projects.

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Monitoring and Logging

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PRMLO-001/002/003

- Provide customer visibility on DevOps approach via continuous monitoring
- Track activity and API usage, enable analyzing/debugging applications
- Leverage Amazon CloudWatch Insights, AWS VPC Flow logs, AWS X-Ray, and AWS CloudTrail logs examples

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PRMLO-001

Good Example vs Unacceptable/Insufficient Information example

"At AnyCompany, continuous monitoring is a Dev Ops practice implemented for our customers. We have a specific approach to identify, observe and detect issues and threats during each phase of the DevOps pipeline. Our ***tool sets include Amazon CloudWatch, CloudTrail, VPC Flow Logs,Config rules, GuardDuty, IAM Analyzer, Security Hub, KMS logging and third-party services DataDog.*** *We work with customers to customize this based on the maturity level of their continuous monitoring practice (****see attached link for AWS health metric, logs, and other tools*** *used for the associated case study). "*

A simple description on monitoring without dashboard or snapshots of how it's used for customers is not acceptable.

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PRMLO-002

Good Example vs Unacceptable/Insufficient Information example

*"We leverage **AWS X-Ray** to analyze and debug the production applications across microservices such as Lambda Functions and Step functions via AWS CDK. Additionally, we work with customers to incorporate **distributed tracing to their preferred tools such as Splunk, Dynatrace if needed.** See attached **a customer example** where we used a combination of both. (Partner attached snapshots of specific customer example)."*

A simple description on monitoring without dashboard or snapshots of how it's used for customer examples is not acceptable.

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PRMLO-003

Good Example vs Unacceptable/Insufficient Information example

"Any Company has a mature methodology and practices for tracking and monitoring applications/APIs. We deploy ***CloudTrail in every AWS Account we manage via Landing Zone or Control Tower*** *to monitor API Activity for Multi-Tenant services. All Logs are stored in a central AWS Account where it can be analyzed with* ***QuickSight****. The CloudTrail Logs are also replicated into a CloudWatch log group and all activities are integrated into* ***GuardDuty and Amazon Inspector.*** *Findings are push to our centralized event management tool, allowing notification and elevation notices to customer stakeholder teams. We provided some details/snapshops of CloudTrail events and API requests related with two customer examples (partners attached snapshots). "*

A simple description on API usage tracking without dashboard or snapshots of how it's used for customer examples is not acceptable.

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Platform as a Service

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PRPAS-001

Orchestration to Run and Manage Web Apps

- AWS Partners leverage orchestration tools as part of DevOps strategy for PaaS

- Refer to AWS Elastic Beanstalk Sample Applications

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PRPAS-001

Good Example vs Unacceptable/Insufficient Information example

"AnyCompany understands the benefits of PaaS like AWS Elastic Beanstalk where we can orchestrate Web application deployment with just a few clicks. This can free developers from deployment-oriented tasks, such as provisioning servers, setting up load balancing, or managing scaling, especially for customers who don't have lots of Cloud Computing expertise. We ***choose AWS Elastic Beanstalk for all PoC work*** *and also for production workloads if it supports the application platform and environment variables required for the customer application. See customer example A for more details. For scenarios where AWS Elastic Beanstalk is not suited for the customer business and/or complex architecture requirements, we* ***use CDK to build templates/platforms for more customized control****s on application deployment or server configurations. See customer example A for more details..."*

This is not applicable to the specific DevOps projects of customer example.

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Security

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PRSEC-001

Communication of DevSecOps Best Practices

- Understanding AWS Security processes and technologies is a shared responsibility

- Incorporate AWS security and data encryption best practices

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PRSEC-001

Documentation Example

How can you implement this?

- Best Practices for Security, Identity, & Compliance
- AWS Well-Architected Framework – Security

Make sure your evidence has:

- Educational documents around security
- “Shift-left” of security practices encouragement
- Details on how encryption is implemented at all layers of the application
- Description of tools, services, softwares and code leveraged

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PRSEC-001

Good Example vs Unacceptable/Insufficient Information example

*"We incorporate **security frameworks with a shift left approach**, adding activities along the customer engagement lifecycle and through automation for seamless integration of security with the business. From the AWS perspective we implement all kinds of approaches to protect information and mitigate possible damage, from strong identity foundation, enabling traceability, applying security to all layers, automating security and protecting data in transit and at rest with AWS services and best practices configuration. From the consulting perspective we have also developed **AWS Well Architected framework review** where we cover the security pillar as guidelines and best practices on the cloud adoption journey. We mainly help organizations to adopt security practices by designing secure applications and integrating security into the architecture and infrastructure design. Attached is our security on-boarding document for customers."*

(partner attached links or documents)

… …"

A simple description or response without PDF, screenshot, or attached documents that contains the Security best practices is not acceptable.

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Common Requirements

Covered in earlier enablement sessions.

Please listen in to any of these recordings to go over the Common Requirements segment

Recording of the Amazon Redshift webinar
OR
Recording of the AWS Lambda webinar

AWS SDP(Service Delivery Program) Common Controls - Deck

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Resources

- ✓ Competency & SDP Common Customer Example Requirements Calibration Guide
- ✓ DevOps Competency Validation Checklist
- ✓ DevOps Competency Calibration Guide
- ✓ Specialization Program Application Guide
- ✓ Program Benefits Guide

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Thank you!

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